

IMDB Classification (Sentiment Analysis)

By Abdullah

Libraries

```
In [1]: # !pip install tensorflow
```

```
In [56]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from tensorflow.keras.preprocessing.text import text_to_word_sequence
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences

from tensorflow.keras import models
from tensorflow.keras import layers
from tensorflow.keras import losses
from tensorflow.keras import metrics
from tensorflow.keras import optimizers
```

```
In [2]: ## =====> Libraries for Preprocessing

import re
import string
import nltk
nltk.download('all')
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem.porter import PorterStemmer
from nltk.stem import WordNetLemmatizer
from nltk.corpus import wordnet
import unicodedata
import html
# stop_words = stopwords.words('english')
```

```
In [101... from sklearn.metrics import accuracy_score, classification_report
```

Reading Dataset

```
In [58]: from pathlib import Path
import os
DATA_PATH=Path('./dat/')
DATA_PATH.mkdir(exist_ok=True)
if not os.path.exists('./dat/aclImdb'):
    !curl -O http://ai.stanford.edu/~amaas/data/sentiment/aclImdb_v1.tar.gz
    !tar -xf aclImdb_v1.tar.gz -C {DATA_PATH}
```

```
In [59]: CLASSES = ['neg', 'pos']  
PATH=Path('./dat/aclImdb/')  
  
def get_texts(path):  
    texts, labels = [], []  
    for idx, label in enumerate(CLASSES):  
        for fname in (path/label).glob('*.*'):  
            texts.append(fname.open('r', encoding='utf-8').read())  
            labels.append(idx)  
    return texts, labels
```

```
In [60]: trn_texts, trn_labels = get_texts(PATH/'train')  
tst_texts, tst_labels = get_texts(PATH/'test')
```

```
In [61]: for i in trn_texts[:15]:  
    print(i, end='\n\n')
```

Predictable parody, just about failed to impress throughout it's looooooong eight minutes. The only thing that made it worthwhile was the DO NOT COLORIZE line at the end credits. Shame something more entertaining wasn't put on the DVD, like Jonathan Ross' enjoyable profile of Romero on 'The Incredible Strange Film Show.'

Painfully bad Christmas film that has an equally painfully bad performance by Vince Vaughn, who is paying his usual frat boy self but this time for a children's movie but without the wit or charm that is in his R rated films. Vaughn seems like he's on autopilot though most of the film and he keeps running into walls with his lackluster performance. After 30 minutes into the film, you would be in touch with your inner scourge and say "Bahumbag" at how unfunny this film is and after another 30 minutes, you will want to walk out over how unbearable the film has gotten during that point. Out of all the actors involved in this mess, only Paul Giamatti and Rachel Weisz brings some life to their perspective paper thin roles and that's mainly because they are both way too good of actors to be in this film. Paul Giamatti brings some depth and warmth to the character of Saint Nick himself but he's forced to Vaughn's level of juvenile behavior when they are doing their sad sibling infighting. You can see in Giamatti's face that he's not having fun with his role and it painfully shows in certain parts of the film. Rachel Weisz brings a sense of fun and spirit to her role but she really does not have much of a character to work with and you can see in her face that she's well aware of that, so much so that she seems irritated in certain parts of the film. Fortunately for her, she's not in the film much at all and is able to save some face, unlike Giamatti, who looks like he's about to fire his agent by the end of the movie. The direction also feels uninspiring, like there is no feeling or flow to be had and this is supposed to be a holiday movie but it ends up feeling like you are just staring at a fancy widow display that is being torn down.

I don't know what went wrong here but with only two actors involved (Giamatti and Weisz) trying their best to at least bring something to the table with a unfunny script they had to work with, spotty direction with no feeling for the subject at hand (and this is a Christmas movie of all things) and a actor who just does not care about his performance (Vaughn), you have a very unevenly bad film that is very painful to watch.

I love Dracula but this movie was a complete disappointment! I remember Lee from other Dracula films from when i was younger, and i thought he was great, but this movie was really bad. I don't know if it was my youth that fooled me into believing Lee was the ultimate Dracula, with style, looks, attraction and the evil underneath that. Or maybe it was just this film that disappointed me.

But can you imagine Dracula with an snobbish English accent and the body language to go along with it? Do you like when a plot contains unrealistic choices by the characters and is boring and lacks any kind of tension..? Then this is a movie for you!

Otherwise - don't see it! I only gave it a 2 because somehow i managed to stay awake during the whole movie.

Sorry but if you liked this movie then you must have been sleep deprived and home alone in a dark room with lots of unwatched space behind you. Maybe alone in your parents house or in a strangers home. Cause not even the characters in this flick seemed afraid, and i think that sums up the whole thing!

Or maybe you like this film because of it's place in Dracula cinema history, perhaps being fascinated by how the Dracula story has evolved from Nosferatu to what it is today. Cause as movie it isn't that appealing, it doesn't pull you in to the suggestive mystery that for me makes the Vampyre myth so fascinating.

And furthermore it has so much of that tacky 70ies feel about it. The scenery looks like cheap Theatre. And i don't say that rejecting everything made in the 70ies. Cause i can love old film as well as new.

Start of with the good bit: several times Swayze talks Zulu to his friends or that language is heard among the tribes. That's a great plus, as normally USA & UK movie audiences think all people on this planet speak English (just in case you're one of them: no they don't).

But the acting is 'tenenkrommend' as we s

ay in The Netherlands (it makes your toes curl -and not in a good way). I like Swayze but in this he's awful. The muscles in his jaws make overtime and he's frowning the whole movie -some one must have told him it looks butch. No Patrick: it looks silly and is compensation for lack of character. Alison Doody (Elizabeth) has opted for a style of acting that does not meet the style of her co-workers. Her acting is so relaxed that this movie could have been set in the current days. And it's not. Your frock was a clue, Alison.

The best acting came from the people from the African Continent and Sided Onyulo as Umbopa I liked best. Clear, warm and in character, his performance is a joy to watch.

General: it is mwah-entertaining on a rainy day. Pity. Could have been better. Sack the director.

This was a very brief episode that appeared in one of the "Night Gallery" show back in 1971. The episode starred Sue Lyon (of Lolita movie fame) and Joseph Campanella who play a baby sitter and a vampire, respectively. The vampire hires a baby sitter to watch his child (which appears to be some kind of werewolf or monster) while he goes out at night for blood. The baby sitter is totally oblivious to the vampire's appearance when she first sees him and only starts to put two and two together when she notices that he has no reflection in the mirror, has an odd collection of books in the library on the occult, and hears strange noises while the vampire goes to talk to the child. She realizes that the man who hired her may not be what she thought he was originally. She bolts out the door, the vampire comes out looking puzzled and the episode is over. I don't know what purpose it was to make such an abbreviated episode that lasted just 5 minutes. They should just have expanded the earlier episode by those same 5 minutes and skipped this one. A total wasted effort.

The title alone (along with the poster) is enough to give away "The Projected Man" as an obvious rip-off of "The Fly". And Bryant Haliday, while much better than the typical IMDb review would have you think, is nobody's idea of an acceptable stand-in for Vincent Price. Although, come to think of it, who would be, unless Michael Gough was available?? Still, if you are in the mood to watch a British "Hammer" style movie with a science fiction theme about a teleportation experiment gone horribly wrong...well, you still might want to give "The Projected Man" a pass and rummage around in the 'remaindered' bin at your local Wal-Mart for another teleporter-accident movie. Because this one just isn't all that good.

Haliday caught a lot of good natured ribbing from the MST3K crew for his part in this movie and in "Devil Doll", but he is actually the best thing in TPM. Maybe he can't carry the movie, but he gets practically no help here from the screenplay. The script bogs down any forward momentum the plot may have in a mire of nonsense about funding and university politics and a guy named Lembach and some sinister cabal who want the teleportation machine to fail so they can steal its secrets...or something. So all the dramatic sequences in the first half of movie involve either phone calls or unconvincing special effects with transparent espresso machines and teleporting rats. Then when poor Haliday gets mutilated by his machine, he has to spend the last part of the film wearing a diaper over half his face and rubber cement over the rest while he electrocutes various Londoners who chance across his path. Tom Cruise and Eric Roberts using bullhorns couldn't have made this screenplay work.

Meanwhile all the other actors diligently try to inject life and interest into their roles for this turgid little project, but the screenplay just swallows their efforts whole. The corrupt project administrator frets and fumes and hisses into the phone to his blackmailers, all the while failing to notice that he looks like a werewolf outfitted in a tweed suit and a Tattersall vest. Haliday's research assistant and ex-girlfriend have the least convincing romance in the history of British horror cinema. His secretary is forced to parade around in her "smalls". None of it really works or gels into a real movie. And it all just kind of stops dead, leaving the viewer going, "Eh? excuse me, wasn't there supposed to be an ENDING here??"

Still, for all its problems, I can easily name a dozen horror movies from the same period that were as bad or worse, and so could anyone else who follows movies (or who has ever browsed the IMDb

b "Bottom 100"). I wouldn't actually pay money to own "Projected Man", but if it were included in some compilation along with a dozen other movies in a DVD collection, I'd probably feel OK about having it. It's a harmless diversion, perfect for a horror movie film festival, to watched with friends while consuming many beers and snacks on a Saturday evening.

Richard Dreyfuss is, indeed, in this flick, but in a rather small part. He is NOT the "obsessed" filmmaker - he's the group's business manager/accountant. Even the box describes the film inaccurately. There are no erotic scenes with Sondra Locke, as advertised, unless one uses the term "erotic" quite loosely. I would not have considered viewing the film without Richard Dreyfuss being in it as a major character. I might have, however, had I realized that the famous 60's anthem, Leonard Cohen's "Suzanne," was an artistic influence. Other than the brief recitation of lines from the end of James Joyce's "Ulysses", and an interesting visual reference to the end of Ingmar Bergman's "The Seventh Seal," I found it a poor attempt to meld symbolic elements and moods immortalized in films like "Last Year at Marienbad" and "Un Chien Andalou." If you like the idea of the eccentric artistic trope, there are many superior films, ranging from "Bye, Bye, Brasil" to "Cecil B. Demented."

My interest in Dorothy Stratten caused me to purchase this video. Although it had great actors/actresses, there were just too many subplots going on to retain interest. Plus it just wasn't that interesting. Dialogue was stiff and confusing and the story just flipped around too much to be believable. I was pretty disappointed in what I believe was one of Audrey Hepburn's last movies. I'll always love John Ritter best in slapstick. He was just too pathetic here.

I don't what that other review was talking about. This definitely isn't a bimbo movie, in fact, I don't think there was one decent looking girl in it. No, it's just cheesy, poorly-done, Ed Wood-style science fiction schlock. And it's bad. I can't even begin to tell you how bad it is. I saw it late at night on cable, and I was in shock. The fact that this movie was ever released is an insult to us all. The actors were either friends of the producer or mentally retarded, the special effects are a joke, and the pace is insanely slow. To me though, the music tops it all. A monkey could write a better theme with a toy xylophone. Do not rent this thing, but if you ever see it on cable, watch it. You'll be amazed at how bad a movie can be.

Apparently, The Mutilation Man is about a guy who wanders the land performing shows of self-mutilation as a way of coping with his abusive childhood. I use the word 'apparently' because without listening to a director Andy Copp's commentary (which I didn't have available to me) or reading up on the film prior to watching, viewers won't have a clue what it is about.

Gorehounds and fans of extreme movies may be lured into watching The Mutilation Man with the promise of some harsh scenes of splatter and unsettling real-life footage, but unless they're also fond of pretentious, headache-inducing, experimental art-house cinema, they'll find this one a real chore to sit through.

82 minutes of ugly imagery accompanied by dis-chordant sound, terrible music and incomprehensible dialogue, this mind-numbingly awful drivel is the perfect way to test one's sanity: if you've still got all your marbles, you'll switch this rubbish off and watch something decent instead (I watched the whole thing, but am well aware that I'm completely barking!).

Mimicking its long title the movie finds ways to come close to the 90' mark. The beautiful sets are here with all that made the Hamer production values a trademark, yet Paris drowned in the fog is a sign of indolent neglect. The story is obvious and can be summed up in a dozen words so there comes nothing unexpected and nothing worth more than 5% of your attention to be expected.

The directing is heavy as a direct transfer from the stage play, actors are mostly stiff as wax figures (ok this is a Hamer feature, only it's sometimes better featured in t

he whole package). My conclusion: this movie is trash, not worth the time I spend that evening. Eternal life is a boring matter and I should have hoped the guys in charge of programming at the Cinemathèque would have known better.

"After the atomic bombs carried by a shot-down Soviet bomber explode in the Arctic, the creature 'Gamera' is released from his hibernation. The giant prehistoric turtle proceeds on a path to Tokyo and destroys anything in his path. The military and the scientific community rush to find a means to stop this monster before Tokyo is laid to waste," according to the DVD sleeve's synopsis.

The re-produced for American audiences version of this, the first film in the "Gamera" series, adds English language material that is even funnier than the regularly dubbed Japanese fare. Clearly, the monster is following in the footsteps of "Godzilla". Taking his cue from ABC's faddish "Batman!" TV series, musician Wes Farrel's ludicrous theme song heightens the US version's camp appeal.

*** Gamera the Invincible (12/15/66) Sandy Howard, Noriaki Yuasa ~ Dick O'Neill, Brian Donlevy, Albert Dekker, John Baragrey

The statistics in this movie were well researched. There is no doubt about it! Al Gore certainly presents his case very well and it is no wonder that this movie got the praise that it got. Al Gore is certainly quite an actor. He sounds so concerned. But actions speak louder than words! Throughout this movie, there are political tidbits and references to his political career sprinkled throughout the movie.

Jimmy Carter, unlike Al Gore, is a man of integrity who not only talks the talk, but walks the walk as well. When Carter thought we needed to conserve energy, he turned down the thermostat in the White House and got warm by wearing a sweater.

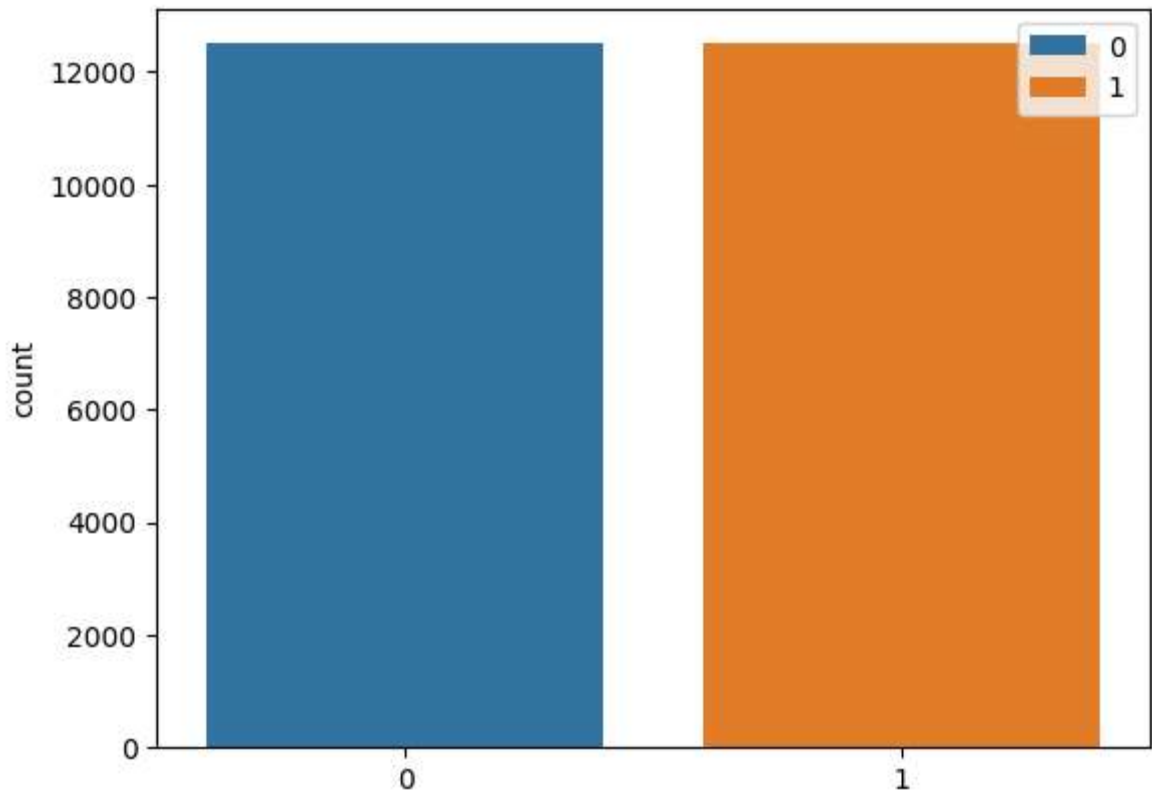
Al Gore tells us that we have to conserve energy and claims that we are creating global warming while he travels around in his own private jet. How much energy does his jet use and how much more pollution does his jet create? How much energy does it take to heat Gore's swimming pool behind his mansion? It would be nice if we could conserve electricity by using smaller appliances and making it a point to turn off anything that is not being used. But if we did, the power company would react to a 50% reduction of energy by calling it a "50% loss in revenue" and recouping their losses by raising the rates by 50%. So "just turning it off" would not be a very good idea.

This movie is a veiled appeal to allow Big Government to take control of everything, in the name of saving planet earth, that is.

This film features two of my favorite guilty pleasures. Sure, the effects are laughable, the story confused, but just watching Hasselhoff in his Knight Rider days is always fun. I especially like the old hotel they used to shoot this in, it added to what little suspense was mustered. Give it a 3.

This movie kinda let me down. It seemed a lot like the movie Jaws when the Hopper was telling the Mayor to close parks was like when Roy Shider was telling the Mayor to close the beaches. They both said no way its summer! But the box says Hopper has to get into the mind of a killer and think like one. But he really doesn't do anything too interesting or exciting. I'm not even a little convinced he and his partner have any experience doing police work when they are in the office wondering how they are gonna solve this case. They just say lets do police work and we'll solve it. And whats up with all the old men with pool cues. I didn't even begin to believe that they were mob bosses. And then the guy who was doubting the guy the mob picked to handle finding the killer. With his hundred dollar haircut and that he thinks his Di@k is the size of a schoolbus. Come on what cruddy lines. I thought he was gonna hit him with a baseball bat like in the other movies. I got this movie used and wouldn't buy it new. I suggest you skip this movie. Oh and it was funny seeing the microphone above the scene where hopper is going out to get coffee.

```
In [62]: sns.countplot(x = trn_labels, hue=trn_labels)
plt.show()
```



Preprocessing

```
In [63]: def remove_special_chars(text):
    re1 = re.compile(r' +')
    x1 = text.lower().replace('#39;', '"').replace('amp;', '&').replace('#146;',
        'nbsp;').replace('#36;', '$').replace('\n', '\n').replace('quot;',
        '<br />', '\n').replace('\\"', '').replace('<unk>', 'u_n').replace(' @.
        ' @-@ ', '-').replace('\ ', ' \ ').replace("I've", "I have").replace("i
    return re1.sub(' ', html.unescape(x1))

def remove_non_ascii(text):
    """Remove non-ASCII characters from list of tokenized words"""
    return unicodedata.normalize('NFKD', text).encode('ascii', 'ignore').decode(

def to_lowercase(text):
    return text.lower()

def remove_punctuation(text):
    """Remove punctuation from list of tokenized words"""
    translator = str.maketrans('', '', string.punctuation)
    return text.translate(translator)

def replace_numbers(text):
    """Replace all interger occurrences in list of tokenized words with textual
    return re.sub(r'\d+', '', text)
```

```

def remove_whitespaces(text):
    return text.strip()

def remove_stopwords(words, stop_words):

    return [word for word in words if word not in stop_words]

# def stem_words(words):
#     """Stem words in text"""
#     stemmer = PorterStemmer()
#     return [stemmer.stem(word) for word in words]

def lemmatize_words(words):
    """Lemmatize words in text"""

    lemmatizer = WordNetLemmatizer()
    return [lemmatizer.lemmatize(word) for word in words]

def lemmatize_verbs(words):
    """Lemmatize verbs in text"""

    lemmatizer = WordNetLemmatizer()
    return ' '.join([lemmatizer.lemmatize(word, pos='v') for word in words])

def text2words(text):
    return word_tokenize(text)

def normalize_text( text):
    text = remove_special_chars(text)
    text = remove_non_ascii(text)
    text = remove_punctuation(text)
    text = to_lowercase(text)
    text = replace_numbers(text)
    words = text2words(text)
    words = remove_stopwords(words, stop_words)
    words = lemmatize_words(words)
    words = lemmatize_verbs(words)

    return ' '.join(words)

```

```

In [64]: def normalize_corpus(corpus):
        return [normalize_text(t) for t in corpus]

```

```

In [65]: trn_texts = normalize_corpus(trn_texts)
        tst_texts = normalize_corpus(tst_texts)

```

```

In [66]: for t in trn_texts[:10]:
        print(t,end='\n\n')

```

predictable parody fail impress throughout loooooong eight minute thing make worthwhile colorize line end credit shame something entertain wasnt put dvd like jonathan ross enjoyable profile romero incredible strange film show

painfully bad christmas film equally painfully bad performance vince vaughn pay usual frat boy self time childrens movie wit charm r rat film vaughn seem like he autopilot though film keep run wall lackluster performance minute film would touch inner scourge say bahumbag unfunny film another minute want walk unbearable film get point actor involve mess paul giamatti rachel weisz bring life perspective paper thin role thats manly way good actor film paul giamatti bring depth warmth character saint nick he force vaughns level juvenile behavior sad sibling infighting see giamatti face he fun role painfully show certain part film rachel weisz bring sense fun spirit role really much character work see face shes well aware much seem irritate certain part film fortunately shes film much able save face unlike giamatti look like he fire agent end movie direction also feel uninspiring like feel flow suppose holiday movie end feel like star fancy widow display tear dont know go wrong two actor involve giamatti weisz try best least bring something table unfunny script work spotty direction feel subject hand christmas movie thing actor care performance vaughn unevenly bad film painful watch

love dracula movie complete disappointment remember lee dracula film younger think great movie really bad dont know youth fool believe lee ultimate dracula style look attraction evil underneath maybe film disappoint imagine dracula snobbish english accent body language go along like plot contain unrealistic choice character bore lack kind tension movie otherwise dont see give somehow manage stay awake whole movie sorry like movie must sleep deprive home alone dark room lot unwatched space behind maybe alone parent house stranger home cause even character flick seem afraid think sum whole thing maybe like film place dracula cinema history perhaps fascinate dracula story evolve nosferatu today cause movie isnt appeal doesnt pull suggestive mystery make vampyre myth fascinate furthermore much tacky y feel scenery look like cheap theatre dont say reject everything make y cause love old film well new

start good bite several time swayze talk zulu friend language hear among tribe thats great plus normally usa uk movie audience think people planet speak english cause youre one dont act tenenkrommend say netherlands make toe curl good way like swayze he awful muscle jaw make overtime he frown whole movie one must tell look butch patrick look silly compensation lack character alison doody elizabeth opt style act meet style coworkers act relax movie could set current day frock clue alison best act come people african continent side onyulo umbopa like best clear warm character performance joy watch general mwahentertaining rainy day pity could better sack director

brief episode appear one night gallery show back episode star sue lyon lolita movie fame joseph campanella play baby sitter vampire respectively vampire hire baby sitter watch child appear kind werewolf monster go night blood baby sitter totally oblivious vampire appearance first see start put two two together notice reflection mirror odd collection book library occult hear strange noise vampire go talk child realize man hire may think originally bolt door vampire come look puzzle episode dont know purpose make abbreviate episode last minute expand earlier episode minute skip one total waste effort

title alone along poster enough give away project man obvious ripoff fly bryant holiday much better typical imdb review would think nobody idea acceptable standing vincent price although come think would unless micheal gough available still mood watch british hammer style movie science fiction theme teleportation experiment go horribly wrongwell still might want give project man pas rummage around remainder bin local walmart another teleporteraccident movie one isnt good holiday catch lot good natured rib mstk crew part movie devil doll actually best thing tpm maybe cant carry movie get practically help screenplay script bog forward momentum pl

ot may mire nonsense fund university politics guy name lembach sinister cabal want teleportation machine fail steal secretsor something dramatic sequence first half movie involve either phone call unconvincing special effect transparent espres so machine teleport rat poor haliday get mutilate machine spend last part film we ar diaper half face rubber cement rest electrocute various londoner chance across path tom cruise eric robert use bullhorn couldnt make screenplay work meanwhile a ctor diligently try inject life interest role turgid little project screenplay sw allow effort whole corrupt project administrator fret fume hiss phone blackmailer fail notice look like werewolf outfit tweed suit tattersall vest halidays research assistant exgirlfriend least convince romance history british horror cinema secretary force parade around small none really work gel real movie kind stop dead leave viewer go eh excuse wasnt suppose end still problem easily name dozen horror movie period bad worse could anyone else follow movie ever browse imdb bottom wouldnt actually pay money project man include compilation along dozen movie dvd collection id probably feel ok harmless diversion perfect horror movie film festival watch friend consume many beer snack saturday even

richard dreyfus indeed flick rather small part obsess filmmaker he group business manageraccountant even box describe film inaccurately erotic scene sonda locke advertise unless one us term erotic quite loosely would consider view film without richard dreyfus major character might however realize famous anthem leonard cohen s suzanne artistic influence brief recitation line end jam joyce ulysses interest visual reference end ingmar bergman seventh seal find poor attempt meld symbolic element mood immortalize film like last year marianbad un chien andalou like idea eccentric artistic troupe many superior film range bye bye brasil cecil b demented

interest dorothy stratten cause purchase video although great actorsactresses many subplots go retain interest plus wasnt interest dialogue stiff confuse story flip around much believable pretty disappoint believe one audrey hepburn last movie ill always love john ritter best slapstick pathetic

dont review talk definitely isnt bimbo movie fact dont think one decent look girl cheesy poorlydone ed woodstyle science fiction schlock bad cant even begin tell bad saw late night cable shock fact movie ever release insult u actor either friend producer mentally retard special effect joke pace insanely slow though music to p monkey could write better theme toy xylophone rent thing ever see cable watch youll amaze bad movie

apparently mutilation man guy wander land perform show selfmutilation way cop abusive childhood use word apparently without listen director andy copps commentary didnt available read film prior watch viewer wont clue gorehounds fan extreme movie may lure watch mutilation man promise harsh scene splatter unsettle reallife footage unless theyre also fond pretentious headacheinducing experimental arthouse cinema theyll find one real chore sit minute ugly imagery accompany dischordant sound terrible music incomprehensible dialogue mindnumbingly awful drivelf perfect way test one sanity youve still get marble youll switch rubbish watch something decent instead watch whole thing well aware im completely bark

Splitting

```
In [67]: from sklearn.model_selection import train_test_split
y = np.array(trn_labels)

def splitting(X):
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    return X_train, X_test, y_train, y_test
```

Model

```
In [68]: from tensorflow.keras.optimizers import AdamW
def ann_model(X_train, X_test, y_train, y_test, epoch=10):
    model = models.Sequential()

    model.add(layers.Dense(20, activation='relu')) # model.add(layers.Dense(32, ac
    model.add(layers.Dropout(0.6))
    model.add(layers.Dense(1, activation='sigmoid'))
    model.summary()
    model.compile(
        optimizer=AdamW(learning_rate=0.005, weight_decay=0.004),
        # optimizer='adam',
        loss=losses.binary_crossentropy,
        metrics=[metrics.binary_accuracy])
    history = model.fit(X_train,
                        y_train,
                        epochs=epoch,
                        batch_size=256,
                        validation_data=(X_test, y_test))
    return history ,model
```

```
In [69]: def result_plot(history):
    history_dict = history.history

    acc = history_dict['binary_accuracy']
    val_acc = history_dict['val_binary_accuracy']
    loss = history_dict['loss']
    val_loss = history_dict['val_loss']
    epochs = range(1, len(acc) + 1)
    plt.clf()

    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(8, 8), sharex=True)

    ax1.plot(epochs, acc, 'bo', label='Training acc')
    ax1.plot(epochs, val_acc, 'b', label='Validation acc')
    ax1.set_title('Training and Validation Accuracy')
    ax1.set_ylabel('Accuracy') # Corrected from 'Loss' to 'Accuracy'
    ax1.legend()
    ax1.grid(True)

    ax2.plot(epochs, loss, 'bo', label='Training loss')
    ax2.plot(epochs, val_loss, 'b', label='Validation loss')
    ax2.set_title('Training and Validation Loss')
    ax2.set_xlabel('Epochs')
    ax2.set_ylabel('Loss')
    ax2.legend()
    ax2.grid(True)

    plt.tight_layout()
    plt.show()
```

Use the diff ways to Representation using Tensorflow & Keras


```
In [70]: # texts = trn_texts + tst_texts
        texts = trn_texts
```

```
In [71]: vocab_sz = 20000 # None means all
        tok = Tokenizer(num_words=vocab_sz, oov_token='UNK')
        tok.fit_on_texts(texts)
```

```
In [72]: len(tok.word_index)
```

```
Out[72]: 91356
```

1. Count

```
In [73]: bow1 = tok.texts_to_matrix(trn_texts , mode='count')
        test_text_count = tok.texts_to_matrix(tst_texts , mode='count')
        bow1[1]
```

```
Out[73]: array([0., 2., 4., ..., 0., 0., 0.])
```

```
In [74]: type(bow1), type(test_text_count)
```

```
Out[74]: (numpy.ndarray, numpy.ndarray)
```

```
In [75]: X_train_count, X_test_count, y_train_count, y_test_count = splitting(bow1)
```

```
In [76]: type(X_train_count), type(X_test_count), type(y_train_count), len(y_test_count)
```

```
Out[76]: (numpy.ndarray, numpy.ndarray, numpy.ndarray, 5000)
```

```
In [77]: history_count, model_coun = ann_model(X_train_count, X_test_count, y_train_count
```

Model: "sequential_5"

Layer (type)	Output Shape	Param #
dense_10 (Dense)	?	0 (unbuilt)
dropout_3 (Dropout)	?	0
dense_11 (Dense)	?	0 (unbuilt)



Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

Non-trainable params: 0 (0.00 B)

Epoch 1/15
79/79  **5s** 48ms/step - binary_accuracy: 0.7686 - loss: 0.4877
 - val_binary_accuracy: 0.8836 - val_loss: 0.2874

Epoch 2/15
79/79  **3s** 39ms/step - binary_accuracy: 0.8956 - loss: 0.2684
 - val_binary_accuracy: 0.8868 - val_loss: 0.2838

Epoch 3/15
79/79  **3s** 37ms/step - binary_accuracy: 0.9213 - loss: 0.2099
 - val_binary_accuracy: 0.8878 - val_loss: 0.2895

Epoch 4/15
79/79  **3s** 35ms/step - binary_accuracy: 0.9323 - loss: 0.1720
 - val_binary_accuracy: 0.8892 - val_loss: 0.3022

Epoch 5/15
79/79  **3s** 35ms/step - binary_accuracy: 0.9456 - loss: 0.1425
 - val_binary_accuracy: 0.8864 - val_loss: 0.3234

Epoch 6/15
79/79  **3s** 35ms/step - binary_accuracy: 0.9503 - loss: 0.1308
 - val_binary_accuracy: 0.8840 - val_loss: 0.3558

Epoch 7/15
79/79  **3s** 36ms/step - binary_accuracy: 0.9537 - loss: 0.1114
 - val_binary_accuracy: 0.8836 - val_loss: 0.3749

Epoch 8/15
79/79  **3s** 38ms/step - binary_accuracy: 0.9575 - loss: 0.1057
 - val_binary_accuracy: 0.8848 - val_loss: 0.3988

Epoch 9/15
79/79  **3s** 34ms/step - binary_accuracy: 0.9616 - loss: 0.0969
 - val_binary_accuracy: 0.8836 - val_loss: 0.4125

Epoch 10/15
79/79  **3s** 33ms/step - binary_accuracy: 0.9623 - loss: 0.0915
 - val_binary_accuracy: 0.8822 - val_loss: 0.4371

Epoch 11/15
79/79  **3s** 33ms/step - binary_accuracy: 0.9643 - loss: 0.0872
 - val_binary_accuracy: 0.8846 - val_loss: 0.4649

Epoch 12/15
79/79  **3s** 33ms/step - binary_accuracy: 0.9681 - loss: 0.0788
 - val_binary_accuracy: 0.8844 - val_loss: 0.4839

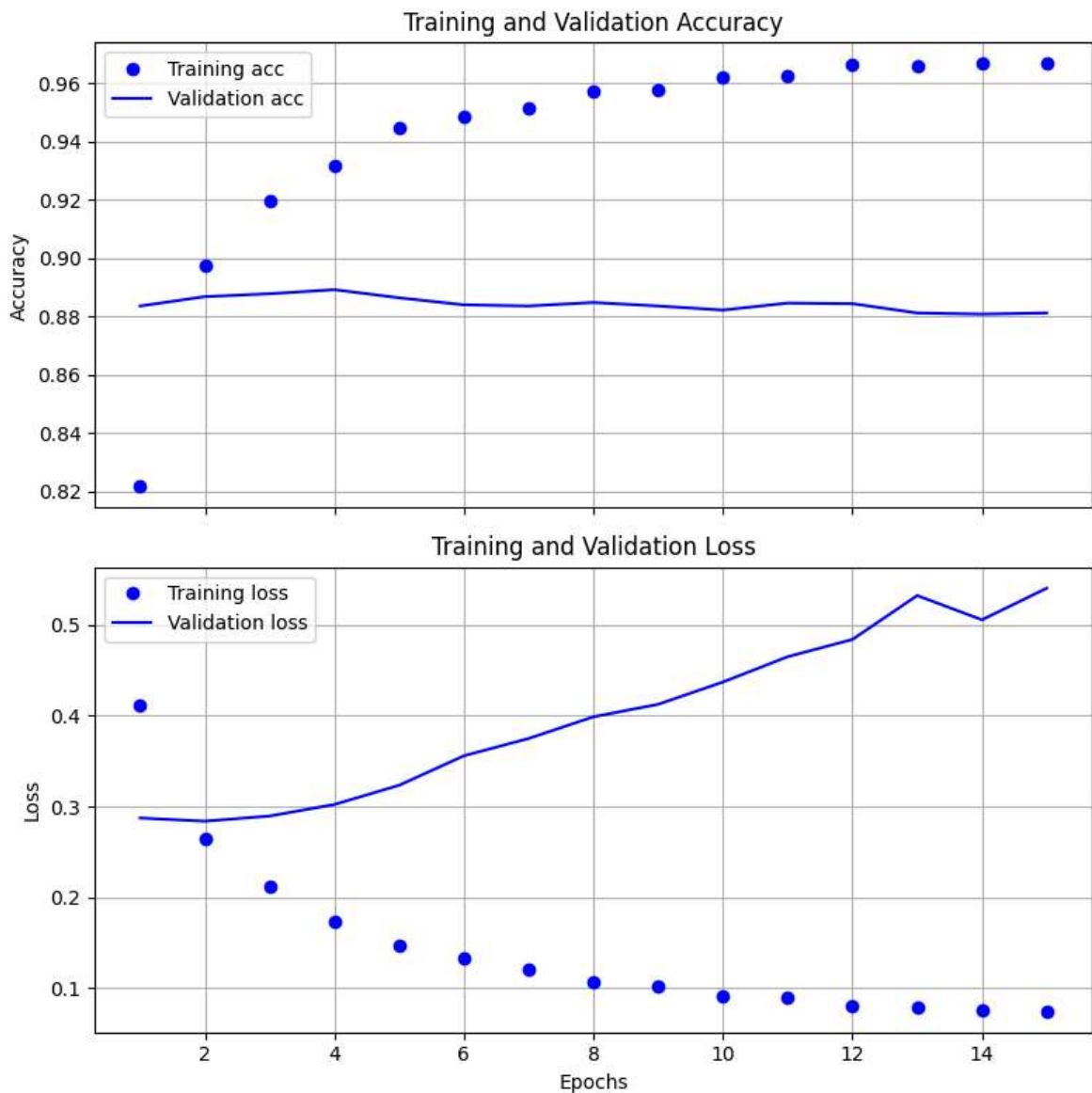
Epoch 13/15
79/79  **3s** 32ms/step - binary_accuracy: 0.9660 - loss: 0.0809
 - val_binary_accuracy: 0.8812 - val_loss: 0.5323

Epoch 14/15
79/79  **3s** 35ms/step - binary_accuracy: 0.9676 - loss: 0.0756
 - val_binary_accuracy: 0.8808 - val_loss: 0.5056

Epoch 15/15
79/79  **3s** 35ms/step - binary_accuracy: 0.9660 - loss: 0.0765
 - val_binary_accuracy: 0.8812 - val_loss: 0.5404

In [78]: `result_plot(history_count)`

<Figure size 640x480 with 0 Axes>



```
In [79]: y_pred_count = model_coun.predict(test_text_count)
```

782/782 ————— 1s 1ms/step

```
In [80]: test_text_count.shape
```

```
Out[80]: (25000, 20000)
```

```
In [81]: (y_pred_count.shape)
```

```
Out[81]: (25000, 1)
```

```
In [82]: binary_predictions = (y_pred_count >= 0.5).astype(int)
```

```
# Flatten if needed (removes extra dimensions)
binary_predictions = binary_predictions.flatten()

print(binary_predictions)
```

```
[0 0 0 ... 1 1 1]
```

```
In [83]: from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [84]: accuracy_score(tst_labels, binary_predictions)
```

Out[84]: 0.86088

In [85]: `confusion_matrix(tst_labels, binary_predictions)`

Out[85]: `array([[10714, 1786],
[1692, 10808]])`

In [103... `print(classification_report(tst_labels, binary_predictions))`

```

              precision    recall  f1-score   support

     0       0.83         0.89         0.86         12500
     1       0.88         0.82         0.85         12500

 accuracy          0.85          0.85          0.85         25000
 macro avg         0.85          0.85          0.85         25000
 weighted avg      0.85          0.85          0.85         25000

```

Freq

In [86]: `bow2 = tok.texts_to_matrix(trn_texts , mode='freq')`
`test_text_freq = tok.texts_to_matrix(tst_texts , mode='freq')`
`bow2[9]`

Out[86]: `array([0. , 0.02020202, 0.01010101, ..., 0. , 0. ,
0.])`

In [87]: `X_train_freq, X_test_freq, y_train_freq, y_test_freq = splitting(bow2)`

In [88]: `history_freq, model_freq = ann_model(X_train_freq, X_test_freq, y_train_freq, y_`

Model: "sequential_6"

Layer (type)	Output Shape	Param #
dense_12 (Dense)	?	0 (unbuilt)
dropout_4 (Dropout)	?	0
dense_13 (Dense)	?	0 (unbuilt)



Total params: 0 (0.00 B)

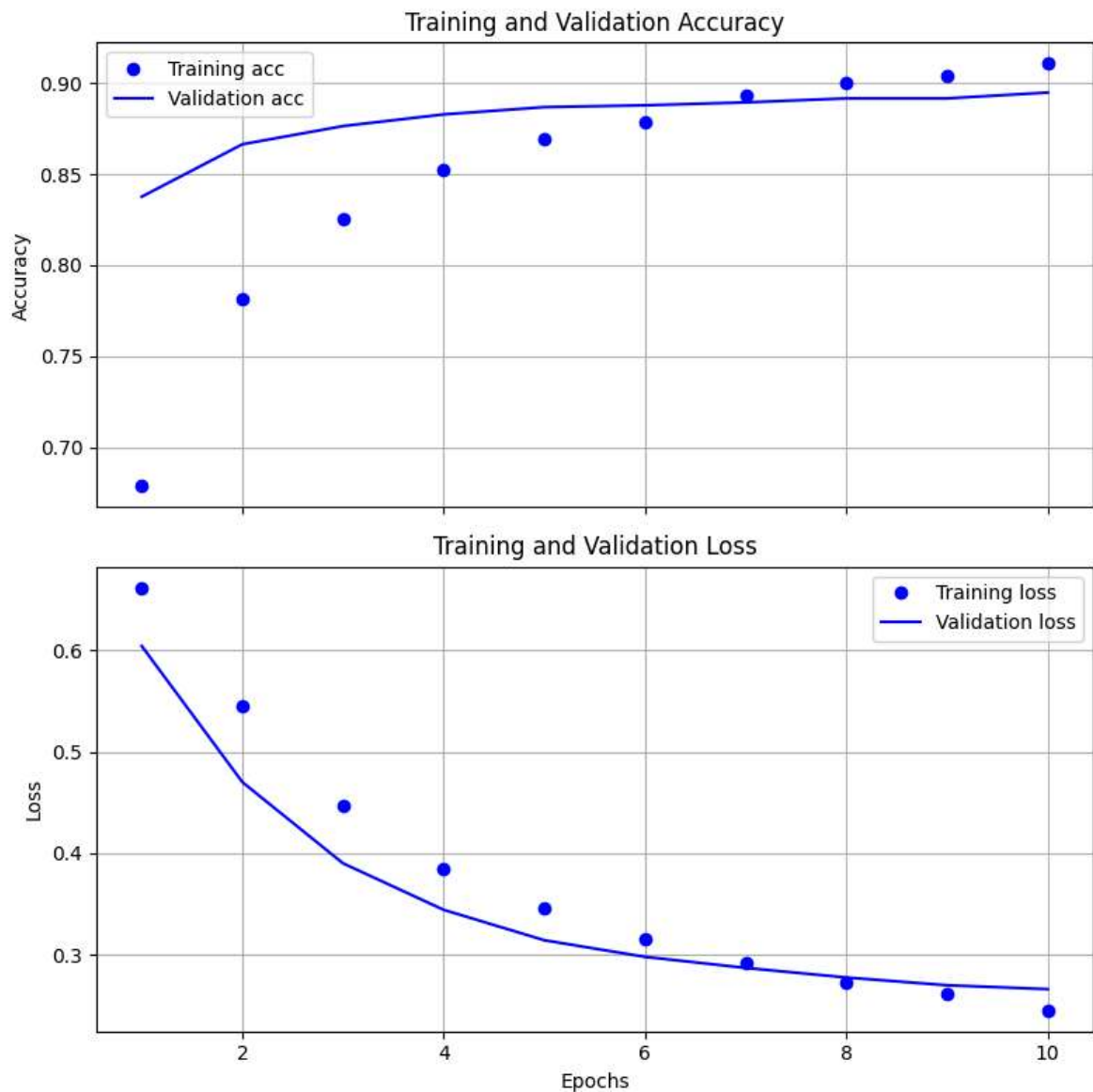
Trainable params: 0 (0.00 B)

Non-trainable params: 0 (0.00 B)

Epoch 1/10
79/79  5s 43ms/step - binary_accuracy: 0.6179 - loss: 0.6801
- val_binary_accuracy: 0.8376 - val_loss: 0.6044
Epoch 2/10
79/79  3s 35ms/step - binary_accuracy: 0.7700 - loss: 0.5733
- val_binary_accuracy: 0.8664 - val_loss: 0.4702
Epoch 3/10
79/79  3s 34ms/step - binary_accuracy: 0.8193 - loss: 0.4619
- val_binary_accuracy: 0.8764 - val_loss: 0.3902
Epoch 4/10
79/79  3s 34ms/step - binary_accuracy: 0.8497 - loss: 0.3931
- val_binary_accuracy: 0.8828 - val_loss: 0.3444
Epoch 5/10
79/79  3s 35ms/step - binary_accuracy: 0.8665 - loss: 0.3557
- val_binary_accuracy: 0.8868 - val_loss: 0.3144
Epoch 6/10
79/79  3s 33ms/step - binary_accuracy: 0.8771 - loss: 0.3197
- val_binary_accuracy: 0.8878 - val_loss: 0.2980
Epoch 7/10
79/79  3s 32ms/step - binary_accuracy: 0.8948 - loss: 0.2912
- val_binary_accuracy: 0.8894 - val_loss: 0.2871
Epoch 8/10
79/79  3s 31ms/step - binary_accuracy: 0.9002 - loss: 0.2777
- val_binary_accuracy: 0.8916 - val_loss: 0.2777
Epoch 9/10
79/79  3s 35ms/step - binary_accuracy: 0.9031 - loss: 0.2667
- val_binary_accuracy: 0.8916 - val_loss: 0.2700
Epoch 10/10
79/79  3s 35ms/step - binary_accuracy: 0.9111 - loss: 0.2459
- val_binary_accuracy: 0.8948 - val_loss: 0.2662

In [89]: `result_plot(history_freq)`

<Figure size 640x480 with 0 Axes>



```
In [90]: y_pred_freq = model_freq.predict(test_text_freq)
```

782/782 ————— 1s 2ms/step

```
In [91]: binary_predictions = (y_pred_freq >= 0.5).astype(int)
         binary_predictions_freq = binary_predictions.flatten()
```

```
In [92]: accuracy_score(tst_labels, binary_predictions_freq)
```

Out[92]: 0.8816

```
In [93]: confusion_matrix(tst_labels, binary_predictions_freq)
```

Out[93]: array([[11177, 1323],
[1637, 10863]])

```
In [104... print(classification_report(tst_labels, binary_predictions_freq))
```

	precision	recall	f1-score	support
0	0.87	0.89	0.88	12500
1	0.89	0.87	0.88	12500
accuracy			0.88	25000
macro avg	0.88	0.88	0.88	25000
weighted avg	0.88	0.88	0.88	25000

TFIDF

```
In [94]: bow3 = tok.texts_to_matrix(trn_texts , mode='tfidf')
test_text_tfidf= tok.texts_to_matrix(tst_texts , mode='tfidf')
bow3[9]
```

```
Out[94]: array([ 0.          , 17.14594463,  0.9377181 , ...,  0.          ,
                0.          ,  0.          ])
```

```
In [95]: X_train_tfidf, X_test_tfidf, y_train_tfidf, y_test_tfidf = splitting(bow3)
```

```
In [96]: history_tfidf, model_tfidf= ann_model(X_train_tfidf, X_test_tfidf, y_train_tfidf)
```

Model: "sequential_7"

Layer (type)	Output Shape	Param #
dense_14 (Dense)	?	0 (unbuilt)
dropout_5 (Dropout)	?	0
dense_15 (Dense)	?	0 (unbuilt)



Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

Non-trainable params: 0 (0.00 B)

Epoch 1/20
79/79  5s 45ms/step - binary_accuracy: 0.7604 - loss: 0.4970
- val_binary_accuracy: 0.8826 - val_loss: 0.2927

Epoch 2/20
79/79  3s 35ms/step - binary_accuracy: 0.9057 - loss: 0.2590
- val_binary_accuracy: 0.8830 - val_loss: 0.2956

Epoch 3/20
79/79  3s 32ms/step - binary_accuracy: 0.9262 - loss: 0.2015
- val_binary_accuracy: 0.8814 - val_loss: 0.3188

Epoch 4/20
79/79  3s 35ms/step - binary_accuracy: 0.9365 - loss: 0.1659
- val_binary_accuracy: 0.8784 - val_loss: 0.3416

Epoch 5/20
79/79  3s 32ms/step - binary_accuracy: 0.9459 - loss: 0.1432
- val_binary_accuracy: 0.8792 - val_loss: 0.3631

Epoch 6/20
79/79  3s 31ms/step - binary_accuracy: 0.9490 - loss: 0.1279
- val_binary_accuracy: 0.8770 - val_loss: 0.3899

Epoch 7/20
79/79  3s 34ms/step - binary_accuracy: 0.9604 - loss: 0.1050
- val_binary_accuracy: 0.8738 - val_loss: 0.4189

Epoch 8/20
79/79  3s 35ms/step - binary_accuracy: 0.9625 - loss: 0.0948
- val_binary_accuracy: 0.8782 - val_loss: 0.4545

Epoch 9/20
79/79  3s 35ms/step - binary_accuracy: 0.9639 - loss: 0.0897
- val_binary_accuracy: 0.8782 - val_loss: 0.5107

Epoch 10/20
79/79  3s 40ms/step - binary_accuracy: 0.9638 - loss: 0.0937
- val_binary_accuracy: 0.8796 - val_loss: 0.5123

Epoch 11/20
79/79  3s 33ms/step - binary_accuracy: 0.9675 - loss: 0.0827
- val_binary_accuracy: 0.8810 - val_loss: 0.5459

Epoch 12/20
79/79  3s 34ms/step - binary_accuracy: 0.9666 - loss: 0.0836
- val_binary_accuracy: 0.8800 - val_loss: 0.5669

Epoch 13/20
79/79  3s 35ms/step - binary_accuracy: 0.9711 - loss: 0.0720
- val_binary_accuracy: 0.8792 - val_loss: 0.5991

Epoch 14/20
79/79  3s 35ms/step - binary_accuracy: 0.9708 - loss: 0.0741
- val_binary_accuracy: 0.8752 - val_loss: 0.5960

Epoch 15/20
79/79  3s 33ms/step - binary_accuracy: 0.9716 - loss: 0.0696
- val_binary_accuracy: 0.8798 - val_loss: 0.6434

Epoch 16/20
79/79  3s 35ms/step - binary_accuracy: 0.9684 - loss: 0.0757
- val_binary_accuracy: 0.8818 - val_loss: 0.6196

Epoch 17/20
79/79  3s 35ms/step - binary_accuracy: 0.9733 - loss: 0.0670
- val_binary_accuracy: 0.8816 - val_loss: 0.6942

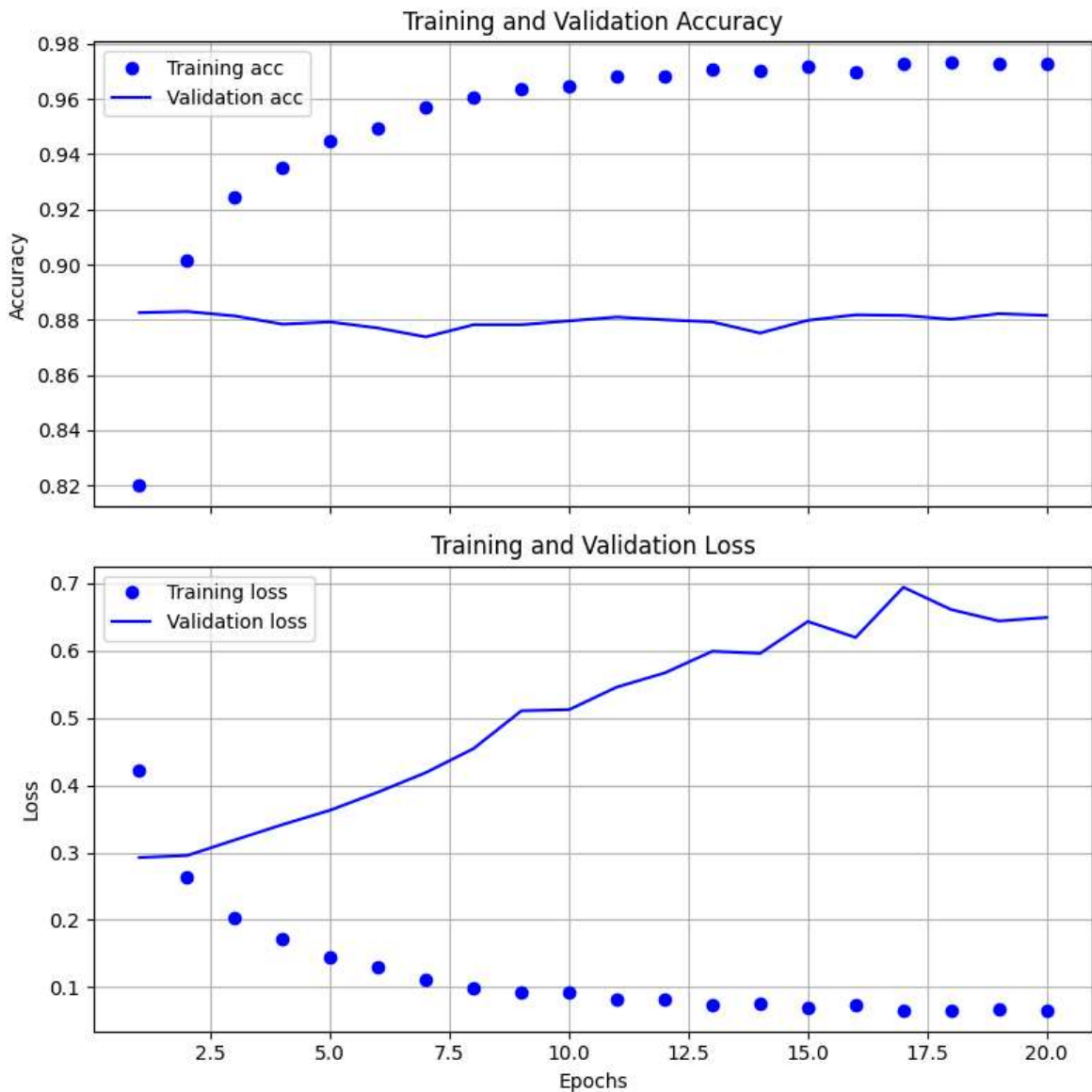
Epoch 18/20
79/79  3s 34ms/step - binary_accuracy: 0.9747 - loss: 0.0634
- val_binary_accuracy: 0.8802 - val_loss: 0.6610

Epoch 19/20
79/79  3s 35ms/step - binary_accuracy: 0.9737 - loss: 0.0636
- val_binary_accuracy: 0.8822 - val_loss: 0.6441

Epoch 20/20
79/79  3s 34ms/step - binary_accuracy: 0.9730 - loss: 0.0642
- val_binary_accuracy: 0.8816 - val_loss: 0.6493


```
In [97]: result_plot(history_tfidf)
```

<Figure size 640x480 with 0 Axes>



```
In [98]: y_pred_tfidf = model_tfidf.predict(test_text_tfidf)
```

782/782 ————— 1s 2ms/step

```
In [99]: binary_predictions = (y_pred_tfidf >= 0.5).astype(int)
         binary_predictions_tfidf = binary_predictions.flatten()
         print(binary_predictions)
```

```
[[0]
 [0]
 [0]
 ...
 [1]
 [1]
 [1]]
```

```
In [100... accuracy_score(tst_labels, binary_predictions_tfidf)
```

```
Out[100... 0.85332
```

```
In [105... print(classification_report(tst_labels, binary_predictions_tfidf))
```

	precision	recall	f1-score	support
0	0.83	0.89	0.86	12500
1	0.88	0.82	0.85	12500
accuracy			0.85	25000
macro avg	0.85	0.85	0.85	25000
weighted avg	0.85	0.85	0.85	25000

In our experiments, we observed significant differences in performance among the three text representation methods Count, Frequency, and TF-IDF. Frequency consistently outperformed the others, it's avoiding over fitting. These results highlight the importance of choosing an appropriate representation method, as it can greatly influence classification performance depending on the nature of the dataset. Also at another project can found another way is the best.

Thanks